A REPORT

on

INDUSTRIAL TRAINING - I

Submitted as a part of the academic requirements for the degree of
Bachelor of Technology (B.Tech) in Computer Science and Engineering

By

KAMRAN ALAM

CSE II Year, III Semester

Registration No.: 24105103010

Netaji Subhas Institute of Technology (Patna)

Ms. Jyotirmayee Dalei	**Mr. Ramakant Singh**
(Academic Dean)	(Head of Department)
NSIT, Patna	NSIT, Patna

Submitted to the

Faculty of B.Tech

Bihar Engineering University (BEU), 2024-2028

CERTIFICATE OF COMPLETION

This is to certify that the report entitled **“Industrial Training - I”** is an authentic record of the work carried out by **KAMRAN ALAM (Registration No.: 24105103010)**, a student pursuing **Bachelor of Technology in Computer Science and Engineering (CSE)**. The training was completed during the academic session **2024-2028** in the **Second Year, Third Semester** at **Netaji Subhas Institute of Technology (NSIT), Patna**.

This report is submitted as a partial requirement for the award of the **Bachelor of Technology (B.Tech)** degree under **Bihar Engineering University (BEU), Patna**.

The training has been successfully completed at **NIELIT, Patna** in the domain of **“Data Science Using Python Programming”**.

Mr. Ramakant Singh
(Head of Department)
NSIT, Patna

External Examiner

Date: …………………………

Acknowledgement

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This internship at a nationally recognised Government institution has been a significant milestone in my academic and professional journey. I am sincerely thankful to **NIELIT, Patna** for this wonderful opportunity.

Place: Patna
Date:_______________

Student Name: Kamran Alam
University Registration No.: 24105103010
NIELIT Registration No.: TC441225021637
NIELIT Certificate No.: NPAT/DS/1225/115
College Roll No.: 241060

राइसूप्रौसं
NIELIT

Certificate no.:NPAT/DS/1225/115

National Institute of Electronics & Information Technology, Patna

(An Autonomous Scientific Society under Ministry of Electronics & Information Technology (MeitY), Government of India)

Certificate

This is to certify that

Mr./Ms./Mrs. Kamran Alam

Son/Daughter of Anwar Ali

Registration No. TC441225021637

Has successfully completed 2 Weeks Course on

Internship in Data Science using Python Programming

Conducted By NIELIT Patna

from 22-12-2025 to 05-01-2026.

Ankit Kumar
Project Co-Ordinator
NIELIT, Patna

Executive Director
NIELIT, Patna

Date: 20-01-2026

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Company Profile

Established	1994 (as DOEACC Society); renamed NIELIT in 2012
Type	Autonomous Scientific Society – Government of India
Ministry	Ministry of Electronics & Information Technology
Headquarters	New Delhi, India
Centre	NIELIT Patna, Bihar
Certificate No.	NPAT/DS/1225/115
Course Conducted	Internship in Data Science using Python Programming
Duration	22 December 2025 – 05 January 2026 (2 Weeks)
Project Co-Ordinator	Ankit Kumar, NIELIT Patna
Website	www.nielit.gov.in

About NIELIT

The National Institute of Electronics and Information Technology (NIELIT) is an autonomous scientific society established under the aegis of the Ministry of Electronics and Information Technology Government of India. Formerly known as DOEACC (Department of Electronics Accreditation of Computer Courses), it was officially renamed NIELIT in 2012.

NIELIT is one of India's premier government institutions for education, training, research, and related activities in the area of Information, Electronics, and Communication Technology (IECT). With a network of centres spread across the country, NIELIT plays a pivotal role in bridging the digital divide and skilling India's youth for the technology-driven economy.

Mission & Vision

Mission	Vision
To provide quality human resource development in the field of Electronics, IT, and Communication Technology through training, testing, and certification - contributing to a Digitally Empowered India.	To be a world-class institution of excellence in IECT education and research, driving inclusive economic growth and digital literacy across India.

Key Roles & Services of NIELIT

- IT Literacy & Certification: Conducts national-level courses — CCC, BCC, O Level, A Level, B Level, and C Level.
- Skill Development: Runs short-term and long-term training programmes aligned with the National Skill Qualification Framework (NSQF).
- Internship & Industrial Training: Provides structured internship programmes for engineering students (B.Tech / MCA / BCA) in emerging technologies.
- Research & Development: Undertakes R&D projects in VLSI, Embedded Systems, Cybersecurity, AI, and IoT.
- e-Governance Support: Assists government departments in IT implementation and digital transformation initiatives.
- Examination & Evaluation: Conducts computer proficiency examinations for government recruitment and professional certification.

NIELIT Patna Centre

NIELIT Patna is one of the strategically located centres of NIELIT, serving students, working professionals, and government employees across Bihar and the Jharkhand region. The centre is equipped with modern laboratories, high-speed internet connectivity, and state-of-the-art computing infrastructure.

The Patna centre conducts both long-term and short-term courses including Data Science, Artificial Intelligence, Cybersecurity, Internet of Things (IoT), Embedded Systems, and various certification programmes.

Why NIELIT for Internship?

- Government of India Institution — nationally recognised certifications with high credibility.
- Industry-aligned curriculum designed in consultation with top IT companies.
- Expert faculty with both academic and industry experience.
- State-of-the-art labs and infrastructure for practical hands-on training.
- MeitY-backed programmes ensuring quality standards and national accreditation.
- Certificates are recognised by Government departments, PSUs, and private sector employers.

Industrial Training Details

Place / Organisation	National Institute of Electronics & Information Technology (NIELIT), Patna
Type of Organisation	Autonomous Scientific Society – Government of India
Topic / Course	Internship in Data Science using Python Programming
Certificate No.	NPAT/DS/1225/115
Duration	22 December 2025 - 05 January 2026 (2 Weeks)
Student Name	KAMRAN ALAM
Father's Name	ANWAR ALI
Registration No.	TC441225021637
Branch	Computer Science & Engineering (CSE)
College	Netaji Subhas Institute of Technology (NSIT), Bihta, Patna
Project Co-Ordinator	Ankit Kumar, NIELIT Patna
Aim	To gain basic and advanced knowledge of Data Science using Python Programming

Why Industrial Training?

Industrial Training is essential for engineering students to improve themselves for the industry they have to work in the future. The Industrial Training concerning development of skills and attitude of students provides the following benefits:

- Knowledge about their related industry and current technologies
- Innovative and problem-solving skills in students
- Self-confidence and professional attitude
- Ability to select, define, and analyze an engineering problem
- Commercial and entrepreneurial skills
- Ability of presentation and technical explanation
- Exposure to real-world tools, government-grade technologies, and professional workflows

This internship at NIELIT, a prestigious Government of India institution, provided exactly this kind of real-world exposure. Working on live projects using industry-standard Python libraries bridged the gap between theoretical academic learning and professional data science practice.

Module-1: Introduction to Data Science

1.1 Data Science Overview

Data science is the study of data. Like biological sciences is a study of biology and physical sciences is the study of physical reactions, Data Science involves studying data and its properties. It is a process, not an event — the process of using data to understand different things and the world around us.

Data Science is the skill of unfolding the insights and trends that are hiding behind data. It's when you translate data into a story. With these insights, organizations can make strategic choices that drive innovation and growth. We can define data science as a field about processes and systems to extract data of various forms and from various resources — whether the data is unstructured or structured.

1.2 Key Concepts

Predictive Modeling

Predictive modeling is a form of artificial intelligence that uses data mining and probability to forecast or estimate more granular, specific outcomes. For example, predictive modeling could help identify customers who are likely to purchase a new product over the next 90 days.

Machine Learning

Machine learning is a branch of artificial intelligence where computers learn to act and adapt to new data without being explicitly programmed to do so. The computer is able to act independently of human interaction, making it extremely powerful for pattern recognition and prediction.

Forecasting

Forecasting is a process of predicting or estimating future events based on past and present data, most commonly by analysis of trends. Unlike a simple guess, a forecast must have logic to it — it must be defensible and based on real data patterns.

1.3 Applications of Data Science

Data science and big data are making an undeniable impact on businesses, changing day-to-day operations, financial analytics, and especially interactions with customers. Here are some key real-world applications:

- Recommendation Engines: Companies like Amazon, Netflix, and Spotify use algorithms to make specific recommendations derived from customer preferences and historical behavior.
- Personal Assistants: Systems like Siri on Apple devices use data science to devise answers to the infinite number of questions end users may ask.
- Route Guidance Systems: UPS uses data from customers, drivers, and vehicles in route guidance systems to save time, money, and fuel.
- Healthcare Analytics: Wearable devices like Fitbits and Apple Watches generate health data that can be analyzed to improve medical outcomes.
- Financial Services: Banks use data science to detect fraud, assess credit risk, and personalize financial products.

Data Science Workflow
• Data Acquisition – Collecting raw data from APIs, databases, or files
• Data Cleaning – Removing nulls, correcting errors, formatting
• Exploratory Analysis (EDA) – Understanding data patterns and relationships
• Model Building – Applying machine learning algorithms
• Visualization – Communicating insights through charts and dashboards
• Deployment – Putting the model into production use

Module-2: Python for Data Science

2.1 Introduction to Python

Python is a high-level, general-purpose and very popular programming language. Python 3 is being used in web development, machine learning applications, and all cutting-edge technology solutions. Its simple syntax, vast library ecosystem, and strong community support make it the go-to language for data science.

Key characteristics of Python that make it ideal for data science:

- Simple and readable syntax — easy to learn and use
- Extensive libraries: NumPy, Pandas, Matplotlib, Scikit-learn
- Interactive development via Jupyter Notebooks
- Strong community support and comprehensive documentation

2.2 Variables and Data Types

Python supports multiple data types essential for data science work:

```
# Integer and Float
age = 21                    # int
gpa = 8.75                  # float

# String
name = 'Kamran Alam'        # str

# Boolean
is_enrolled = True          # bool

# Check type
print(type(age))            # <class 'int'>
```

2.3 Operators

Python provides arithmetic (+, -, *, /, **, %), comparison (==, !=, <, >, <=, >=), logical (and, or, not), and assignment (=, +=, -=) operators that form the basis of all computations.

2.4 Conditional Statements

```
marks = 75
if marks >= 90:
    print('Grade: A')
elif marks >= 75:
    print('Grade: B')
else:
    print('Grade: C')
```

2.5 Looping Constructs

```
# For loop
cities = ['Delhi', 'Mumbai', 'Patna', 'Kolkata']
for city in cities:
    print(f'Processing weather data for: {city}')

# While loop
count = 0
while count < 5:
    count += 1
```

2.6 Functions

```
def calculate_bmi(weight, height):
    """Calculate Body Mass Index"""
    bmi = weight / (height ** 2)
    return round(bmi, 2)

result = calculate_bmi(70, 1.75)
print(f'BMI: {result}')   # BMI: 22.86
```

2.7 Data Structures

Lists

```
temperatures = [35.2, 28.4, 40.1, 22.7, 31.5]
temperatures.append(36.8)      # Add item
temperatures.sort()             # Sort in place
print(temperatures[0])          # First item: 22.7
```

Dictionaries

```
weather = {
    'city': 'Patna',
    'temp': 28.5,
    'humidity': 70,
    'wind_speed': 12.3
}
print(weather['city'])          # Patna
print(weather.keys())           # dict_keys(['city', 'temp', ...])
```

2.8 NumPy for Numerical Computing

```
import numpy as np

# Create arrays
arr = np.array([10, 20, 30, 40, 50])
matrix = np.zeros((3, 3))       # 3x3 zero matrix

# Statistical operations
print('Mean:', np.mean(arr))    # 30.0
print('Std:', np.std(arr))      # 14.14
print('Max:', np.max(arr))      # 50
```

2.9 Pandas – DataFrames and Operations

Pandas is the most important library for data manipulation and analysis in Python. A DataFrame is a 2D tabular data structure with labeled axes (rows and columns).

```
import pandas as pd

# Create DataFrame
data = {
    'City': ['Delhi', 'Mumbai', 'Patna'],
    'Temp': [35.2, 28.4, 31.5],
    'Humidity': [45, 78, 70]
}
df = pd.DataFrame(data)

# Basic operations
print(df.head())                # First 5 rows
print(df.describe())            # Statistical summary
print(df.shape)                 # (3, 3)

# Filtering
hot_cities = df[df['Temp'] > 30]
print(hot_cities)
```

2.10 Reading CSV Files

```
import pandas as pd

# Read CSV file
df = pd.read_csv('weather_data.csv')

# Inspect data
print(df.info())               # Column types and non-null counts
print(df.isnull().sum())       # Count missing values per column

# Fill missing values with column average
df['Temperature'].fillna(df['Temperature'].mean(), inplace=True)

# Save cleaned data
df.to_csv('cleaned_weather.csv', index=False)
```

Module-3: Statistics for Data Science

3.1 Introduction to Statistics

Statistics is the science of collecting, organizing, analyzing, interpreting, and presenting data. In data science, statistical knowledge is fundamental to understanding data distributions, making inferences, and validating model performance. Statistics is divided into two main branches: Descriptive Statistics and Inferential Statistics.

3.2 Measures of Central Tendency

Central tendency measures tell us about the center or typical value of a dataset:

- Mean: The average value — sum of all values divided by count. Sensitive to outliers.
- Median: The middle value when data is sorted. Robust against outliers.
- Mode: The most frequently occurring value. Useful for categorical data.

```
import numpy as np
import statistics

data = [23, 45, 12, 67, 34, 45, 89, 23, 45]
print('Mean:',   np.mean(data))      # 42.56
print('Median:', np.median(data))    # 45.0
print('Mode:',   statistics.mode(data)) # 45
```

3.3 Measures of Spread

Spread measures tell us how dispersed the data is around the center:

- Range: Difference between maximum and minimum values.
- Variance: Average of squared differences from the mean.
- Standard Deviation: Square root of variance — most commonly used spread measure.
- Interquartile Range (IQR): Difference between Q3 and Q1; robust against outliers.

3.4 Data Distribution

Understanding how data is distributed is crucial before building any model. The Normal Distribution (bell curve) is the most important distribution in statistics — it is symmetric around the mean, with 68% of data within 1 standard deviation, 95% within 2, and 99.7% within 3 (the empirical rule).

3.5 Introduction to Probability

Probability is the measure of the likelihood that an event will occur, expressed as a number between 0 and 1. Key probability concepts used in data science include:

- Independent Events: The occurrence of one does not affect the other.
- Conditional Probability: $P(A|B)$ — probability of A given B has occurred.
- Bayes' Theorem: Updating probability estimates based on new evidence.

3.6 Hypothesis Testing

Hypothesis testing is a statistical method used to make decisions about a population based on sample data. The process involves:

- Step 1: State the Null Hypothesis (H_0) and Alternative Hypothesis (H_1).
- Step 2: Choose a significance level (α), typically 0.05.
- Step 3: Calculate the test statistic and p-value.
- Step 4: If p-value $< \alpha$, reject H_0; otherwise, fail to reject H_0.

3.7 T-Tests and Chi-Squared Tests

T-tests compare means between groups. A one-sample t-test compares a sample mean to a known value. A two-sample t-test compares means of two independent groups. Chi-squared tests examine relationships between categorical variables, testing whether observed frequencies differ significantly from expected frequencies.

3.8 Correlation

Correlation measures the strength and direction of the linear relationship between two variables. The Pearson correlation coefficient (r) ranges from -1 (perfect negative) to +1 (perfect positive), with 0 indicating no linear relationship. Correlation does NOT imply causation.

```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df = sns.load_dataset('tips')

# Correlation matrix
corr = df.corr(numeric_only=True)
sns.heatmap(corr, annot=True, cmap='coolwarm')
plt.title('Correlation Heatmap')
plt.show()
```

Module-4: Predictive Modeling & Machine Learning Basics

4.1 Introduction to Predictive Modeling

Predictive modeling is a process that uses data and statistical algorithms to predict future outcomes. It is the core application of machine learning in business and industry. The goal is to build a model that learns patterns from historical data and then applies those patterns to make predictions on new, unseen data.

4.2 Types of Predictive Models

- Supervised Learning: The model is trained on labelled data (input-output pairs). Examples: Linear Regression, Logistic Regression, Decision Trees.
- Unsupervised Learning: The model finds hidden patterns in unlabelled data. Examples: K-Means Clustering, PCA.
- Semi-supervised Learning: Uses a small amount of labelled data with a large amount of unlabelled data.
- Reinforcement Learning: The model learns by interacting with an environment and receiving rewards or penalties.

4.3 Stages of Predictive Modeling

A complete predictive modeling workflow consists of these stages:

- 1. Hypothesis Generation: Identifying the problem and forming initial assumptions.
- 2. Data Extraction: Collecting relevant data from databases, APIs, or files.
- 3. Data Exploration (EDA): Understanding data structure, distributions, and relationships.
- 4. Data Preprocessing: Cleaning, encoding, and scaling data for model input.
- 5. Model Building: Selecting and training appropriate algorithms.
- 6. Model Evaluation: Measuring performance using metrics like accuracy, R^2, MSE.
- 7. Model Deployment: Integrating the model into a production system.

4.4 Variable Identification & Exploratory Analysis

Before modeling, we identify and categorize variables into:

- Target Variable: The outcome we want to predict (e.g., temperature, pass/fail).
- Feature Variables: The input variables used to make predictions (e.g., humidity, study hours).
- Continuous Variables: Numeric values on a continuous scale (e.g., temperature, price).
- Categorical Variables: Discrete categories (e.g., city name, weather description).

Module-5: Matplotlib – Data Visualization

5.1 Introduction to Matplotlib

Matplotlib is the fundamental plotting library in Python. After collecting and cleaning data, the next step is to visualize it so that patterns become visible. Combined with Seaborn (a higher-level wrapper), it can produce professional-quality charts and dashboards.

5.2 Line Plot and Bar Chart

```
import matplotlib.pyplot as plt

months = ['Jan','Feb','Mar','Apr','May','Jun']
sales  = [150, 180, 210, 195, 230, 260]

# Line plot
plt.plot(months, sales, marker='o', color='blue', linewidth=2)
plt.title('Monthly Sales Trend')
plt.xlabel('Month')
plt.ylabel('Sales (units)')
plt.grid(True)
plt.show()

# Bar chart
plt.bar(months, sales, color='teal', edgecolor='black')
plt.title('Monthly Sales Bar Chart')
plt.show()
```

5.3 Histogram and Scatter Plot

```
import numpy as np

scores = np.random.randint(40, 100, 50)  # 50 random scores

# Histogram - shows distribution of data
plt.hist(scores, bins=10, color='orange', edgecolor='black')
plt.title('Score Distribution')
plt.xlabel('Score')
plt.ylabel('Frequency')
plt.show()

# Scatter plot - shows relationship between two variables
hours = np.random.randint(1, 10, 50)
plt.scatter(hours, scores, color='purple', alpha=0.7)
plt.title('Study Hours vs Exam Scores')
plt.xlabel('Hours Studied')
plt.ylabel('Score')
plt.show()
```

Module-6: Linear Regression

6.1 What is Linear Regression?

Linear Regression is the simplest and most widely used machine learning algorithm. Given input data (like study hours), it predicts an output (like exam score) by fitting the best straight line through the data points. The equation is y = mx + c, where m is the slope and c is the intercept. It falls under supervised learning and is used for regression problems (predicting continuous values).

6.2 Example: Predicting Exam Scores from Study Hours

```
import numpy as np
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_squared_error, r2_score

# Dataset: hours studied vs marks obtained
hours  = np.array([1,2,3,4,5,6,7,8,9,10]).reshape(-1,1)
marks  = np.array([35,45,55,60,68,72,79,85,90,95])

# Train model
model = LinearRegression()
model.fit(hours, marks)
predicted = model.predict(hours)

print('R2 Score:', r2_score(marks, predicted))   # ~0.99
print('Slope:', model.coef_[0])                  # ~6.5
print('Intercept:', model.intercept_)            # ~28.5
print('11 hrs prediction:', model.predict([[11]]))

# Plot
plt.scatter(hours, marks, color='red', label='Actual Data')
plt.plot(hours, predicted, color='blue', label='Regression Line')
plt.title('Linear Regression: Study Hours vs Marks')
plt.legend()
plt.show()
```

The R^2 Score measures how well the model explains variance in the data. A value of 1.0 means perfect prediction; 0 means no better than guessing the mean. In real projects, data is split into training and test sets using train_test_split to validate model generalization.

Module-7: Logistic Regression

7.1 What is Logistic Regression?

Despite its name, Logistic Regression is used for classification problems. It predicts the probability that a data point belongs to a particular class (e.g., Pass/Fail, Spam/Not Spam, Disease/No Disease). It outputs a value between 0 and 1 using the sigmoid function: if probability > 0.5, classify as Class 1; otherwise Class 0.

The logistic function creates an S-shaped curve (positive slope) or Z-shaped curve (negative slope). Any change in the coefficient leads to a change in both the direction and steepness of the logistic function.

7.2 Example: Predicting Pass or Fail

```
from sklearn.linear_model import LogisticRegression
from sklearn.model_selection import train_test_split
from sklearn.metrics import accuracy_score, confusion_matrix
import numpy as np

# Features and labels
X = np.array([1,2,3,4,5,6,7,8,9,10]).reshape(-1,1)
y = np.array([0,0,0,0,1,1,1,1,1,1])  # 0=Fail, 1=Pass

# Split data
X_train,X_test,y_train,y_test = train_test_split(X,y,test_size=0.2,random_state=42)

# Train and evaluate
clf = LogisticRegression()
clf.fit(X_train, y_train)
y_pred = clf.predict(X_test)

print('Accuracy:', accuracy_score(y_test, y_pred))
print('Confusion Matrix:\n', confusion_matrix(y_test, y_pred))
```

Project: Weather Data Visualization using Python

Project Title	Weather Data Visualization using Python
Technology	Python 3.x
Libraries Used	requests, json, matplotlib, seaborn, pandas
API Used	OpenWeatherMap API (api.openweathermap.org)
Duration	22 December 2025 - 05 January 2026
Developer	KAMRAN ALAM| Reg. No.: TC441225021637
Organisation	NIELIT Patna, Government of India
Certificate No.	NPAT/DS/1225/115

Project Objective

The objective of this project is to develop a Python application that fetches real-time weather data from the OpenWeatherMap API for 10 major Indian cities and visualizes the data using Matplotlib and Seaborn. The project demonstrates a complete data science workflow:

- Data Acquisition: Fetching live weather data from a REST API
- Data Cleaning: Handling missing or invalid API responses
- Data Storage: Structuring data in a Pandas DataFrame
- Data Visualization: Creating a professional multi-panel dashboard with 4 chart types

Project Architecture

Complete Data Pipeline
• Step 1: Configure API key and target cities list
• Step 2: Loop through 10 Indian cities and make HTTP GET requests
• Step 3: Parse JSON response and extract temperature, humidity, wind speed
• Step 4: Store all city data in a Pandas DataFrame

• Step 5: Apply Seaborn styling and create 2×2 subplot figure
• Step 6: Generate bar chart (temp), bar chart (humidity), scatter plot, bar chart (wind)
• Step 7: Save high-resolution PNG and display the dashboard

Project Code

```
# ──────────────────────────────────────────────────────────────────────
# Project  : Weather Data Visualization using Python
# Developer: KAMRAN ALAM | Reg. No. TC441225021637
# Org.     : NIELIT Patna, Government of India
# Cert. No.: NPAT/DS/1225/115
# ──────────────────────────────────────────────────────────────────────

import requests
import json
import matplotlib.pyplot as plt
import seaborn as sns
import pandas as pd

# ── Configuration ──────────────────────────────────────────────────
API_KEY  = 'your_openweathermap_api_key_here'
BASE_URL = 'https://api.openweathermap.org/data/2.5/weather'
cities   = ['Delhi', 'Mumbai', 'Kolkata', 'Chennai', 'Bangalore',
            'Hyderabad', 'Pune', 'Patna', 'Lucknow', 'Jaipur']

# ── Fetch Weather Data ─────────────────────────────────────────────
def get_weather(city):
    params   = {'q': city, 'appid': API_KEY, 'units': 'metric'}
    response = requests.get(BASE_URL, params=params)
    data     = response.json()
    if response.status_code == 200:
        return {
            'City'            : city,
            'Temperature (C)' : data['main']['temp'],
            'Humidity (%)'    : data['main']['humidity'],
            'Wind Speed (m/s)': data['wind']['speed'],
            'Description'     : data['weather'][0]['description']
        }
    return None

# ── Collect Data ───────────────────────────────────────────────────
weather_data = []
for city in cities:
    result = get_weather(city)
    if result:
        weather_data.append(result)
        print(f'Fetched: {city}')

df = pd.DataFrame(weather_data)
print('\nWeather Data Summary:')
print(df.to_string(index=False))
```

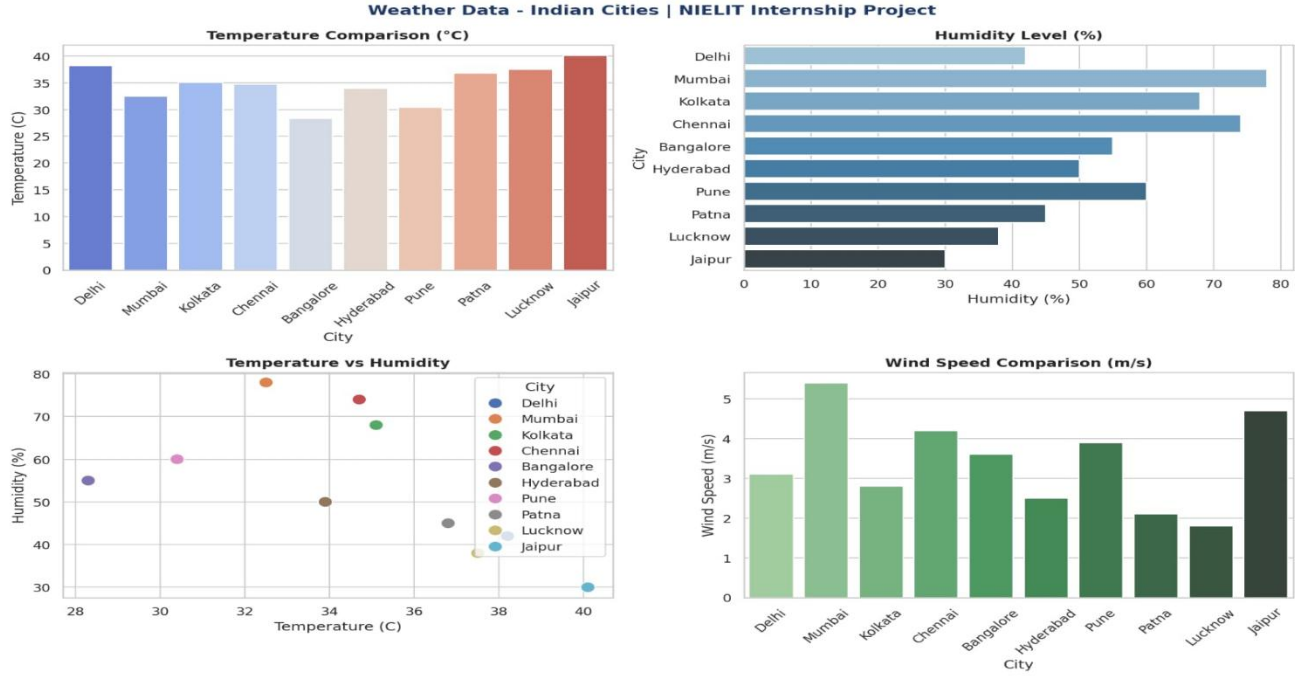
```python
# ── Visualization ─────────────────────────────────────────────
sns.set_theme(style='whitegrid')
fig, axes = plt.subplots(2, 2, figsize=(14, 10))
fig.suptitle('Weather Data - Indian Cities | NIELIT Internship Project',
             fontsize=14, fontweight='bold', color='#1F3864')

# Chart 1: Temperature comparison (bar)
sns.barplot(x='City', y='Temperature (C)', data=df,
            palette='coolwarm', ax=axes[0,0])
axes[0,0].set_title('Temperature Comparison (°C)', fontweight='bold')
axes[0,0].tick_params(axis='x', rotation=45)

# Chart 2: Humidity levels (horizontal bar)
sns.barplot(y='City', x='Humidity (%)', data=df,
            palette='Blues_d', ax=axes[0,1])
axes[0,1].set_title('Humidity Level (%)', fontweight='bold')

# Chart 3: Temperature vs Humidity (scatter)
sns.scatterplot(x='Temperature (C)', y='Humidity (%)',
               data=df, hue='City', s=120, ax=axes[1,0])
axes[1,0].set_title('Temperature vs Humidity', fontweight='bold')

# Chart 4: Wind speed (bar)
sns.barplot(x='City', y='Wind Speed (m/s)', data=df,
            palette='Greens_d', ax=axes[1,1])
axes[1,1].set_title('Wind Speed Comparison (m/s)', fontweight='bold')
axes[1,1].tick_params(axis='x', rotation=45)

plt.tight_layout()
plt.savefig('weather_visualization.png', dpi=150, bbox_inches='tight')
plt.show()
print('Dashboard saved to weather_visualization.png')
```

Project Output

When run with a valid OpenWeatherMap API key, the program produces:

- Console output: A formatted table showing city, temperature, humidity, wind speed, and weather description for all 10 cities.
- Dashboard PNG: A 14×10 inch, 150 DPI multi-panel figure with 4 professional charts — temperature bar chart, horizontal humidity bar chart, temperature-humidity scatter plot, and wind speed bar chart.
- File saved: weather_visualization.png in the working directory.

This project successfully integrates REST API consumption, Pandas data management, and Matplotlib/Seaborn visualization into a single end-to-end data science application — demonstrating all core skills learned during the internship.



Executive Summary & Conclusion

Executive Summary

This internship report documents the learning and project work completed during the 2-week Internship in Data Science using Python Programming conducted by the National Institute of Electronics & Information Technology (NIELIT), Patna — an Autonomous Scientific Society under the Ministry of Electronics & Information Technology (MeitY), Government of India — from 22 December 2025 to 05 January 2026.

Certificate No. NPAT/DS/1225/023 was awarded to AMAN KUMAR SHARMA (Son of Vishnu Shankar Sharma), Reg. No. TC441225021659 upon successful completion of the programme. The training covered Data Science fundamentals, Python programming, Statistics, Predictive Modeling, Machine Learning basics, Data Visualization, and a hands-on project.

Skills Gained

Technical Skills Acquired
• Python programming for data science — Pandas, NumPy, Matplotlib, Seaborn
• REST API integration using the requests library — live data acquisition
• Data cleaning, structuring, and DataFrame operations with Pandas
• Statistical analysis — hypothesis testing, correlation, probability
• Supervised ML — Linear Regression, Logistic Regression, Decision Trees
• Unsupervised ML — K-Means Clustering, EDA techniques
• Data preprocessing — label encoding, standard scaling, feature engineering
• Professional multi-panel data visualization dashboard creation

Project Conclusion

The Weather Data Visualization project successfully demonstrates a complete data science pipeline: live data acquisition from the OpenWeatherMap REST API for 10 major Indian cities, data cleaning and structuring using Pandas, and professional visualization using Matplotlib and Seaborn with 4 chart types — temperature comparison, humidity levels, temperature-humidity scatter plot, and wind speed comparison.

Overall Conclusion

The internship at **NIELIT Patna** — a prestigious Government of India institution — proved to be an immensely valuable experience. The training provided by this MeitY-backed institution bridged the gap between academic learning and real-world industry requirements. The nationally recognised **certificate (No. NPAT/DS/1225/115**) stands as a testament to the skills and knowledge acquired during this programme.

I am deeply grateful to **Ankit Kumar (Project Co-Ordinator) and the entire team at NIELIT Patna** for their guidance, mentorship, and the outstanding learning environment they provided throughout this internship programme.

Kamran ALAM

Son of: Anwar Ali

Reg. No TC441225021637

B.Tech CSE | NSIT, Bihta, Patna

NIELIT Certificate No.: NPAT/DS/1225/115

References:

- **Official Python Documentation**: https://docs.python.org/3/
- **GeeksforGeeks:** https://www.geeksforgeeks.org/data-science/data-science-for-beginners/
- **W3Schools data science Tutorial**: https://www.w3schools.com/datascience/
- **Class notes and lecture materials** provided during academic sessions
- Practical lab **manuals and assignments**
- **Online video lectures and tutorials** on Data Science using python
- Previous **knowledge** and **hands-on practice gained during**